**Solution**

The table for the operation $\times_{11}$ is as follows.

| $\times_{11}$ | 1 | 3 | 4 | 5 | 9 |
|---|---|---|---|---|---|
| 1 | 1 | 3 | 4 | 5 | 9 |
| 3 | 3 | 9 | 1 | 4 | 5 |
| 4 | 4 | 1 | 5 | 9 | 3 |
| 5 | 5 | 4 | 9 | 3 | 1 |
| 9 | 9 | 5 | 3 | 1 | 4 |

Table12.3

Following the same kind of procedure as explained in the previous example, a brief outline of the process of verification of the properties of $\times_{11}$ on A is given below.

(i) Since each box has an unique element of A, $\times_{11}$ is a **binary operation** on A.

(ii) The entries are symmetrical about the main diagonal. Hence $\times_{11}$ has **commutative property.**

(iii) As usual, the **associative property** can be seen to be true.

(iv) The entries of both the row and column headed by the element 1 are identical. Hence 1 is the **identity** element.

(v) Since the identity 1 exists in each row and each column, the **existence of inverse** property is assured for $\times_{11}$. The inverse of 1 is 1, that of 3 is 4, that of 4 is 3, 5 is 9 , and, that of 9 is 5.

## EXERCISE 12.1

1. Determine whether $*$ is a binary operation on the sets given below.

   (i) $a*b = a.|b|$ on $\mathbb{R}$ (ii) $a*b = \min(a,b)$ on $A = \{1,2,3,4,5\}$

   (iii) $(a*b) = a\sqrt{b}$ is binary on $\mathbb{R}$ .

2. On $\mathbb{Z}$, define $*$ by $(m*n) = m^n + n^m : \forall m,n \in \mathbb{Z}$. Is $*$ binary on $\mathbb{Z}$?

3. Let $*$ be defined on $\mathbb{R}$ by $(a*b) = a+b+ab-7$. Is $*$ binary on $\mathbb{R}$? If so, find $3*\left(\frac{-7}{15}\right)$.

4. Let $A = \{a+\sqrt{5}b : a,b \in \mathbb{Z}\}$ . Check whether the usual multiplication is a binary operation on $A$.

5. (i) Define an operation $*$ on $\mathbb{Q}$ as follows: $a*b = \left(\frac{a+b}{2}\right)$; $a,b \in \mathbb{Q}$ . Examine the closure, commutative, and associative properties satisfied by $*$ on $\mathbb{Q}$.

   (ii) Define an operation $*$ on $\mathbb{Q}$ as follows: $a*b = \left(\frac{a+b}{2}\right)$; $a,b \in \mathbb{Q}$ . Examine the existence of identity and the existence of inverse for the operation $*$ on $\mathbb{Q}$.



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